

C UDAY KARTHIK

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TECHNICAL SKILLS

- **Languages** : C/C++, Python, JAVA, JavaScript
- **ML/NLP Tools** : PyTorch, TensorFlow, Scikit-learn, Transformers, Pandas, NumPy, Matplotlib
- **DevOps & Platforms** : Git, GitHub Actions (CI/CD), SQL
- **Web & Data** : MongoDB, Express, Node.js, React, HTML-CSS
- **Tools** : Figma, Excel, MATLAB, RStudio
- **Soft Skills** : Leadership, Team Management, Creative Problem Solving, Communication

EXPERIENCE

- **Freelance Frontend Developer** (Remote | May 2025 – Present)
 - Designed and developed responsive websites for small businesses and individuals, ensuring user-friendly interfaces.
 - Implemented clean, maintainable code using HTML, CSS, and JavaScript.
 - Provided pre-design using Figma.
 - Worked directly with clients to gather requirements, deliver updates, and deploy final products.
 - Provided post-delivery support and minor updates as needed.
- **Leader, BrainStorm Group**
 - Founded and led a student group dedicated to generating and developing unique solutions for real-world problems.
 - Conducted regular brainstorming sessions to identify problems and design viable solutions.
 - Mentored group members, encouraging creative thinking and teamwork.
 - Oversaw project development from idea generation to prototype building and presentation
- **Management Lead – Sports Club**
 - Led event planning and execution, curated insightful knowledge resources, and fostered community engagement.
 - Spearheaded strategic initiatives, making suitable improvements based on previous event feedbacks and also planning unique events to keep the audiences engaged.
 - Led a team of volunteers to ensure smooth event execution and participant engagement.
- **Digital Media Freelancer** (Remote | Feb 2024 – Dec 2024):
 - Created digital content, including graphics, videos, and promotional materials, for local businesses and student groups.
 - Collaborated with clients to understand branding needs and deliver customized solutions on time.

PROJECTS

- **ASL Translator** - (Java, YOLOv5)
 - Created a desktop application to translate American Sign Language gestures into text using **Java**.
 - Integrated image processing and pattern recognition algorithms to capture hand gestures through a webcam and map them to corresponding letters or words.
 - Tested accuracy with multiple gesture datasets to improve recognition performance.
 - Focused on creating an intuitive GUI for user interaction, with real-time translation display.
- **Food Donation WebSite** - (React, Google-Maps, Python, Mongo-DB)
 - Developed a website named Innaikum Karangal using **React** and **Python**.
 - Built a full-stack web platform to bridge the gap between surplus food donors (e.g., restaurants, households) and local NGOs or shelters.
 - Implemented a clean, responsive UI with **React**, backed by **Mongo-DB** to track donations and logistics.
 - Ensured smooth user experience and secure data handling, helping reduce local food wastage through efficient resource redistribution.
 - Designed user authentication and role-based access for donors, NGOs, and administrators.
- **Women Safety System**
 - Developed an AI-powered module that can be attached to any standard security camera to upgrade it into a **smart surveillance system** for women's safety.
 - Focused on creating an affordable, easily deployable add-on solution to enhance existing security infrastructure without needing expensive smart cameras.
 - Designed and trained a Convolutional Neural Network (CNN) model to analyze live video feeds in real time and detect potential threats or suspicious activities.
 - Integrated the module with an **ESP32 microcontroller**, enabling wireless connectivity and IoT functionality for remote alert transmission.
 - Programmed the system to automatically send an **SOS signal** with location details to predefined emergency contacts when danger is detected.
- **Local ChatRoom Application**
 - Designed and developed a secure local network-based chatroom for users within the same LAN/Wi-Fi.
 - Implemented basic encryption for message security and user authentication.
 - Built using Java with socket programming to handle multiple clients and real-time message broadcasting.
 - Created an intuitive front-end for seamless multi-user conversations in real time.

EDUCATION

B.Tech in Artificial Intelligence & Robotics, VIT Chennai (Pursuing) CGPA: 8.16/10 (SEM 1 to 4) 2023-2027

Relevant Coursework : Data Structures and Algorithms, Computer Networks, Database Management, Operating Systems, Probability Theory and Random Processes, Discrete Mathematics.

Grade 12 – 85% 2023

INTERESTS

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| • Management and Events Lead | • Web Development | • Theorising Creative concepts | • Gen-AI | • Volleyball |
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